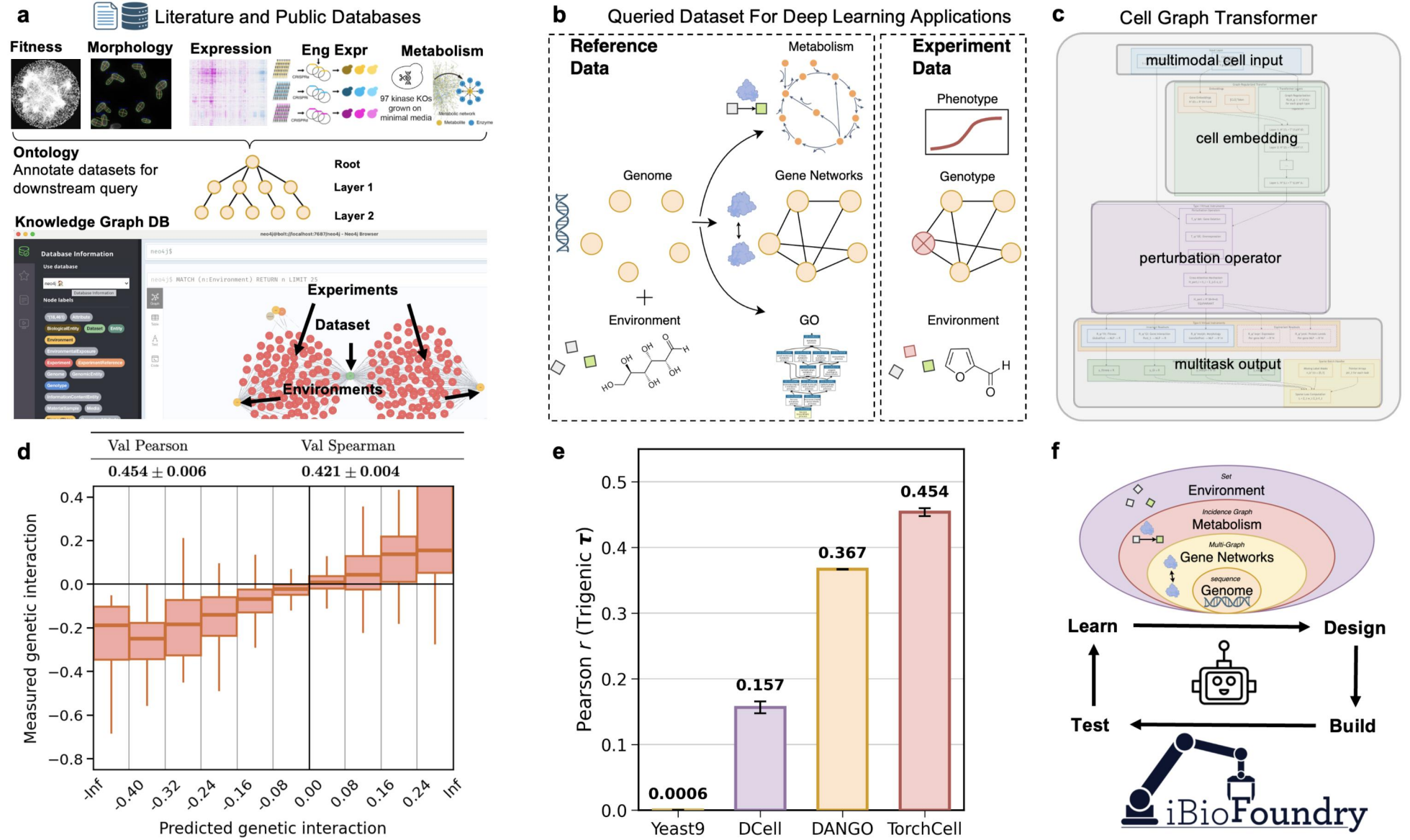

a
Literature and Public Databases
Fitness
Morphology
Expression
Eng Expr
Metabolism
97 kinase KOs grown on minimal media
Ontology
Annotate datasets for downstream query
Root
Layer 1
Layer 2
Knowledge Graph DB
Experiments
Dataset
Environments
b
Queried Dataset For Deep Learning Applications
Reference Data
Experiment Data
Metabolism
Phenotype
Genome
Gene Networks
Genotype
Environment
GO
Environment
c
Cell Graph Transformer
multimodal cell input
cell embedding
perturbation operator
multitask output
d
Val Pearson
0.454 ± 0.006
Val Spearman
0.421 ± 0.004
Measured genetic interaction
Predicted genetic interaction
-Inf
-0.40
-0.32
-0.24
-0.16
-0.08
0.00
0.08
0.16
0.24
Inf
e
Pearson r (Trigenic τ)
0.0006
0.157
0.367
0.454
Yeast9
DCell
DANGO
TorchCell
f
Set
Environment
Incidence Graph
Metabolism
Multi-Graph
Gene Networks
sequence
Genome
Learn
Design
Test
Build
iBioFoundry